Does an LSTM with financial turbulence forecast realised volatility better, and does it matter for the desk?

Strategy research · L/S Gamma Desk, QUANTT

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Question: look into https://github.com/hugogobato/Forecasting-Realized-Volatility-through-Financial-Turbulence-and-Neural-Networks

Method: Read the paper behind the repository (Souto & Moradi 2023, *Economics and Business Review* 9(2), CC BY 4.0; now in `docs/papers/`) and the repository's code and spreadsheets (the Model 2 notebook, the HAR and GARCH workbooks, the data file). Searched all 36 papers in `docs/papers/` (page-tagged index) plus the web. Quotes were verified against the cited PDF pages. Page numbers are PDF pages; printed journal pages are given in brackets.

Previous research: ML and RL for realised volatility forecasting; the desk's 2026-10-18 forecast-ladder card (`research/RV_Forecasting/2026-10-18-forecast-ladder/`).

1 Short answer

The repository is the code for one short study. A single-layer LSTM fed past daily realised volatility (Yang-Zhang) plus Kritzman and Li's financial turbulence index forecasts next-day S&P 500 volatility slightly better than an LSTM without it, HAR and GARCH.

The evidence is thin:

- **One split:** a single 80/20 split of five years of data, so the test set is about 252 days ending November 2022.
- **Small gain:** about 6% lower error than HAR on the S&P 500, and none on the Dow.
- **Mislabelled metric:** the "RMSE" in the code is a mean absolute error.
- **Missing code:** the turbulence series arrives precomputed, so the code that builds it, and whether it looks ahead, can't be checked.

The idea worth keeping is the turbulence index itself. It's a Mahalanobis distance of a day's cross-asset returns, cheap to build from free data, and an independent study found it improves volatility forecasts.

For the desk the forecaster is not the bottleneck. The always-on put book doesn't use the VRP signal, and the 2026-10-18 card found that HAR-X and a random forest don't beat GARCH at 22 days. At most, turbulence earns a cheap forecast-only diagnostic, not an LSTM build.

2 What the papers say

2.1 What the study does

- **Inputs:**
  - **Model 2**, the proposal, takes past RV and turbulence.
  - **Model 1** is an LSTM on RV only.
  - **Model 3** is an LSTM with Bucci's macro inputs.

  - **Models 4-6** are GARCH(1,1), EGARCH(1,1) and HAR.

- **Data:** daily S&P 500 RV over five years, with the Dow and Nasdaq as robustness checks.

  > "daily observations of the S&P 500 realized volatility (RV) from 01 November 2017 until 01 November 2022, five years in total" (Souto & Moradi 2023, p. 10 [140])

  > "The data set is divided into training data, 80% of the total data, and test data, 20% of the total data." (p. 10 [140])

- **Turbulence** is Kritzman and Li's (2010) squared Mahalanobis distance of a day's return vector from its historical mean, $d_t = (y_t - \mu)\Sigma^{-1}(y_t - \mu)'$, with

  > "Σ = sample covariance n × n matrix of historical returns" (p. 5 [135])

  The paper doesn't say which n assets make up the S&P 500 return vector, or over what window μ and Σ are estimated. The appendix says only that turbulence is

  > "estimated with prices retrieved from Compustat" (p. 24 [154])

2.2 What it finds

- **Headline:**

  > "the proposed model yields statistically significantly more accurate and robust forecasts than all other studied models when applied to stocks with middle-to-high volatility" (p. 3 [133])

- **S&P 500 error (Table 4, p. 15 [145]):** 0.318 for Model 2, against 0.339 for the plain LSTM and 0.340 for HAR. GARCH and EGARCH are worst at 0.401 and 0.418, because

  > "they are actually almost always predicting the next time-step value as being the current time-step value" (p. 14 [144])

- **The Dow doesn't support it** (Table 8, p. 19 [149]): HAR 0.282, Model 2 0.283, plain LSTM 0.283. The authors narrow their claim accordingly:

  > "It could be thus hypothesized that Model 2 performs relatively better than the other considered models only when applied stocks with middle-to-high volatility" (p. 18 [148])

- **Limits the authors state:**

  > "the assumption that ten trials were enough to fulfill CLT conditions when estimating the RMSE and MA 95% CI can easily be considered a weakness of this study" (p. 23 [153])

  > "for DM tests, the forecasts of the first trial of the LSTM models will be used" (p. 14 [144])

2.3 What the code shows

These points come from reading the repository's code and spreadsheets, not from the paper:

- **The "RMSE" is a mean absolute error.**
  - The Model 2 notebook computes `sum(sqrt((test - pred)**2)) / n`, and the HAR and GARCH workbooks average `SQRT((actual - predicted)^2)`. Both are a mean absolute error.
  - "Model accuracy" is 1 minus the mean absolute percentage error. The paper describes it as based on MPE:

    > "RMSE and mean percent error (MPE) are used" (p. 13 [143])

  - No loss suited to volatility (QLIKE) is reported, and the 2026-09-27 report and the desk's forecast cards treat QLIKE as the standard.

- **The validation split differs from the paper.**

  > "12.5% of the training data is used as validation data" (p. 10 [140])

  The tuning code uses a 60/20/20 split, so validation is 20% of all data, and it draws about 20 random configurations. The loss list in the code (MSE, MAE, Huber) also differs from the paper's Table 2 (MSE, RMSE, Huber).

- **The forecasts don't peek at test values.** Each step predicts day t from a window ending at t−1, then appends the actual value for t. The scaler is fitted on training data only.

- **The turbulence series can't be checked.** The data file holds a precomputed "S&P 500 Financial Turbulence" column next to RV, and no code builds it. If μ and Σ were estimated over the full five years, test-period information leaks into the training inputs. That can't be ruled out from the repository.

- **The baselines are weak.**

  - GARCH is fitted to daily returns, so it forecasts conditional return volatility, not the Yang-Zhang range measure it is scored against. That puts it at a disadvantage.
  - HAR's coefficients are fixed values in the workbook, fitted once and never refit.
  - No model is refit through the test year.

2.4 Other evidence on turbulence (outside the repo)

- **Kritzman and Li (2010)**, *Financial Analysts Journal* 66(5), defined the index for risk management: stress-testing portfolios and scaling down risk when turbulence is high (CFA Institute).

- **Salisu, Demirer and Gupta (2022)**, *Global Finance Journal* 52, are the independent support. Adding weekly turbulence to a GARCH-MIDAS model improved out-of-sample daily volatility forecasts in seven markets, and turbulence beat a systemic-risk measure (University of Pretoria repository).

- **The same authors' later work** compares newer networks: NBEATSx against LSTM, TCN, HAR and GARCH (2024), and Transformers against HAR and NBEATSx (2024), as listed on Publinova and OUCI. These were seen as listings only, not read.

2.5 What the desk's papers already say about neural RV forecasts

- **HAR is hard to beat.** Neural nets help mostly at longer horizons and with implied vol as an input. From Christensen, Siggaard & Veliyev (`A-machine-learning-approach-to-volatility-forecasting.pdf`), as verified in the 2026-09-27 report:

  > "The random forest and neural networks are preferred among ML algorithms and significantly improve prediction relative to HAR." (p. 36, printed p. 35)

  > "Interestingly, the basic HAR model is remarkably solid and tough to beat in the VaR framework." (p. 33, printed p. 32)

- **Better forecasts don't mean better option P&L.**

  > "Similarly, ML baselines (GBDT, MLP, LSTM, DeepVol, GARCH) struggle with profitability despite occasionally achieving lower volatility" (*OPHR*, p. 9)

3 How the papers connect

Paper	Builds on / agrees with	Differs from	Key contribution
Souto & Moradi 2023 (repo)	Bucci 2020 (LSTM RV); Kritzman & Li 2010; Salisu et al. 2022	Christensen et al. (horizons up to a month, QLIKE, many stocks)	Turbulence as an extra LSTM input for next-day RV
Kritzman & Li 2010 (web)	Mahalanobis (1927)	Volatility-only risk measures	Turbulence index for stress tests and risk scaling
Salisu et al. 2022 (web)	Kritzman & Li	LSTM approach	Turbulence improves GARCH-MIDAS volatility forecasts in 7 markets
Christensen et al. 2022 (repo)	Corsi 2009 (HAR)	Single-index, single-split studies	Random forests and neural nets beat HAR, more so at longer horizons
OPHR 2025 (repo)	Volatility-trading reinforcement learning	Forecast-error-only evaluation	Lower forecast error doesn't mean option P&L

- **Two independent studies agree** that turbulence adds information to volatility forecasts: Salisu et al. with an econometric model, and Souto & Moradi with an LSTM.
- **Souto & Moradi's test is much weaker** than Christensen et al.'s: one index, one test year, absolute-error loss, no refitting.
- **The LSTM part is not where the gain comes from.** The paper shows turbulence helps the LSTM. It doesn't show an LSTM beats a HAR that also gets turbulence (a HAR-X with turbulence), which is the cheap comparison.

4 Relevance to the LS Gamma desk

These are suggestions. The horizon, inputs and any thresholds are the PM's call.

- **The forecaster isn't what drives P&L now.**
  - The always-on stress-sized put book ignores the VRP signal.
  - On the signal book, VRP barely ranked P&L.
  - The 2026-10-18 forecast ladder found HAR-GK, HAR-X and a random forest all lose to GARCH on 22-day QLIKE, mostly because GARCH updates daily through spikes. A better RV forecast is a second-order improvement.
- **The horizon doesn't match.** The paper forecasts one day ahead. The desk's signal needs RV over the option's life, about 22-30 days.
- **If anything is tested, test turbulence, not the LSTM.** A suggested cheap diagnostic, with no backtest trials:
  - **Build the index:** compute turbulence from free daily returns of a small cross-asset set, for example SPY, TLT, GLD, IWM and QQQ, which are also on R2. Use a trailing window for μ and Σ, so nothing looks ahead.
  - **Test it in the forecast-ladder setup:** add the index to HAR-X as one more input, at the 22-day horizon, scored with QLIKE in 2014-2019 and 2020-2024.
  - **Promote only if it earns it:** if it beats GARCH on the ladder's bar, it would earn a P&L card. If not, archive it.

- **As a risk gauge, expect little.** Kritzman and Li meant turbulence for scaling risk. The desk's 2026-11-02 cards showed that sizing by VIX, VVIX or the VIX slope mostly re-times exposure to March 2020. Turbulence will be highly correlated with those, so a sizing card would likely repeat their result. That's an inference, not tested.
- **Not recommended:**
  - building the LSTM itself: it needs many trials and gives a small, fragile gain;
  - using the repository's data: it is S&P 500 index-level Compustat data, not the desk's.

5 Gaps and open questions

- **Turbulence construction:** which assets, which window, and whether μ and Σ use future data. The repository doesn't say.
- **Robustness:** no evidence beyond one test year (late 2021 to late 2022, a falling market) and three US indexes.
- **Losses and benchmarks:** the reported losses aren't suited to volatility, and there's no comparison against HAR plus turbulence or HAR with implied vol.
- **Horizon:** whether turbulence helps 22-day forecasts at all. Salisu et al. report gains at short and long horizons in a different model; the desk would have to check its own horizon.
- **P&L:** none of this connects forecast error to option P&L.

6 Sources

- **Repo papers used:**
  - `Forecasting-Realized-Volatility-through-Financial-Turbulence-and-Neural-Networks.pdf`: Souto & Moradi 2023, pp. 3, 5, 10, 13, 14, 15, 18, 19, 23, 24 (added to `docs/papers/` today).
  - `A-machine-learning-approach-to-volatility-forecasting.pdf`: Christensen, Siggaard & Veliyev, pp. 33, 36.
  - `OPHR-Mastering-Volatility-Trading-with-Multi-Agent-Deep-Reinforcement-Learning.pdf`: p. 9.
- **Repo papers checked, nothing directly relevant:**
  - `Novel-Optimization-Approach-for-Realized-Volatility-Forecast-of-Stock-Price-Index-Based-on-Deep-Reinforcement-Learning-Model.pdf`: an LSTM/GRU/DBN ensemble; it doesn't use turbulence.
  - `Foundation-Time-Series-AI-Model-for-Realized-Volatility-Forecasting.pdf`: a literature mention of LSTM and NBEATSx only.
  - `Improving-S&P-500-Volatility-Forecasting-through-Regime-Switching-Methods.pdf` and `Downside-Risk-Reduction-Using-Regime-Switching-Signals-A-Statistical-Jump-Model-Approach.pdf`: "turbulent" used only in the everyday sense.
  - `Realised-Volatility-Forecasting-Machine-Learning-via-Financial-Word-Embedding.pdf`.
- **Code read:** the GitHub repository `hugogobato/Forecasting-Realized-Volatility-through-Financial-Turbulence-and-Neural-Networks`:
  - the `Model 2/S&P_500.ipynb` notebook;
  - the `Model 4/GARCH model.xlsx` and `Model 6/HAR models.xlsx` workbooks;

  - the `Model 2/S_P 500 Realized Volatility.xlsx` data file.

- **Web:**
  - Souto & Moradi, doi:10.18559/ebr.2023.2.737.
  - Kritzman & Li 2010, CFA Institute.
  - Salisu, Demirer & Gupta 2022, University of Pretoria.
  - Souto & Moradi 2024 listings: NBEATSx, Transformers.

- **Suggested papers to add:**
  - Salisu, Demirer & Gupta 2022: the strongest independent evidence that turbulence helps volatility forecasts.
  - Kritzman & Li 2010: the index's definition and its use in risk management.
  - Souto & Moradi 2024 (NBEATSx): the same authors with more horizons and stronger baselines (LSTM, TCN, HAR, GJR-GARCH).